Gilbert Uwonkunda

Senior Geospatial Solutions Architect · GeoAI & Deep Learning · Urban Intelligence

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PROFESSIONAL PROFILE

Geospatial professional with 5+ years of progressive experience designing and delivering operational AI and enterprise GIS systems for Rwanda's national government. Architect of the Urban Dynamic Map for secondary and satellite cities at MININFRA; trained custom deep learning models for illegal construction detection and forest monitoring; and built a landslide risk scoring application serving the City of Kigali on web and mobile. Experienced in moving solutions from concept to production across urban planning, disaster resilience, climate compliance, and land governance. Trained and consulted for 20+ government institutions across East Africa. Co-founder of Spatial Prime Solutions Ltd, developing AI-powered land intelligence products.

CORE COMPETENCIES

Urban Intelligence	UDM Architecture, Illegal Construction Detection, Building Footprint Extraction, Urban Growth Monitoring
GeoAI & Deep Learning	Custom Model Training, MaskRCNN, ResNet50, PyTorch, ArcGIS Deep Learning, Google Earth Engine
Disaster Resilience	Landslide Risk Scoring, Flood Hazard Analysis, Spatial Risk Modeling, Mobile/Web Risk Applications
Enterprise GIS	ArcGIS Enterprise, ArcGIS Mission, ArcGIS Pro, Survey123, PostgreSQL/PostGIS, SDE, REST APIs
Spatial Engineering	Python (ArcPy, GeoPandas, Rasterio, FastAPI), Satellite Imagery Processing, Workflow Automation
Engagement	Government Stakeholder Reporting, Executive Presentations, Technical Training & Consulting.

FLAGSHIP PROJECTS

Urban Dynamic Map & Spatial Development Framework — Secondary & Satellite Cities *MININFRA · Esri Rwanda*

Designed and deployed the UDM and Spatial Development Framework extending urban planning intelligence to Rwanda's secondary and satellite cities. Provides district-based indicator reporting, map interfaces, and development dashboards to support long-term national urban planning strategy at MININFRA.

Illegal Construction AI Detection Model — City of Kigali *City of Kigali · AI / Deep Learning*

Trained a custom deep learning model for automated illegal construction identification across Kigali. Integrated into the end-to-end AI Building Inspection Hub deployed across 18 districts, with 9 actively onboarded for field operations and permit reconciliation.

Landslide Risk Scoring Application — City of Kigali *City of Kigali · Disaster Risk Reduction*

Developed a per-building landslide risk scoring system for Kigali, delivered as both a mobile and web application. Enables planners and emergency responders to prioritize at-risk structures and support evidence-based disaster preparedness decisions.

Forest Monitoring Deep Learning Model — National Forest Inventory *Government of Rwanda · Climate & Land Use*

Trained a custom model for automated forest extraction and change detection across Rwanda. Monitors nationwide afforestation and deforestation trends to support government climate reporting and landscape management obligations.

Rwanda Green Fund (GCF) Enterprise Tracking Platform *Rwanda Green Fund · ArcGIS Enterprise*

Architected a full-stack ArcGIS Enterprise system for managing and reporting internationally funded green initiatives. Ensures compliance and transparency for GCF-funded projects across Rwanda.

PROFESSIONAL EXPERIENCE

Senior Geospatial Solutions Architect

January 2025 – Present

Esri Rwanda · Kigali, Rwanda

- Lead architect for the City of Kigali AI-powered Building Inspection Hub, integrating NLA and CoK servers via VPN tunnels with automated daily permit synchronization from BPMIS
- Deployed MaskRCNN illegal construction detection model (ResNet50 backbone) applied across 18 districts with 9 secondary(satellite) districts actively onboarded for field inspection workflows.
- Designed complex ArcGIS Enterprise attribute rule workflows covering inspector approval chains, fine management, and automated case status transitions across multiple resolution paths
- Developed Python-based REST API for daily permit reconciliation and produced executive-level dashboards and architectural documentation for City of Kigali district leadership

GIS Specialist

June 2022 – December 2024

Esri Rwanda · Kigali, Rwanda

- **Designed and deployed the** Urban Dynamic Map (UDM) and Spatial Development Framework for secondary and satellite cities at MININFRA,
- providing district-based planning dashboards, indicator reporting, and map interfaces to support Rwanda's national urban development strategy
- **Built a** landslide risk scoring application for the City of Kigali,
- calculating per-building risk scores delivered via mobile and web platforms to support disaster preparedness and urban resilience planning
- **Trained a** custom deep learning model for forest monitoring
- enabling automated change detection for Rwanda's National Forest Inventory and government climate compliance reporting
- Architected the Rwanda Green Fund (GCF) ArcGIS Enterprise platform for tracking and reporting internationally funded climate initiatives
- Deployed a real-time Health Emergency Management System using ArcGIS Mission, synchronizing ambulance response across Kigali and nearby cities
- Developed a centralized mining licensing and inspection portal for end-to-end regulatory lifecycle management at central government level
- Trained and consulted for 20+ government institutions across East Africa on enterprise GIS, spatial analysis, and digital transformation

GIS Consultant
September 2021 – June 2022

PSS Ltd · Kigali, Rwanda

- Delivered spatial analysis, GIS mapping, and custom geospatial workflow development for client projects across multiple sectors

SDG Coordinator
August 2020 – July 2021

UN SDSN Youth · Kigali, Rwanda

- Coordinated SDG youth network programs and supported data collection, stakeholder engagement, and reporting for SDG-aligned initiatives in Rwanda

ENTREPRENEURSHIP

Co-Founder
January 2026 – Present

Spatial Prime Solutions Ltd · Kigali, Rwanda

- Co-founded an early-stage spatial intelligence startup developing AI-powered geospatial products for land governance and urban planning markets
- Architecting a Natural Language Zoning tool enabling users to query land-use regulations through satellite imagery and conversational AI

EDUCATION

Bachelor's Degree in Geography
September 2016 – August 2021

University of Rwanda — College of Science and Technology · Kigali, Rwanda

CERTIFICATIONS

- **Enterprise Geodata Management Professional 2201** — *Esri · January 2025*
- **Esri Young Professionals Network (YPN) Ambassador** — *Esri · May 2024*
- **Generative AI Overview for Project Managers** — *Project Management Institute · June 2024*
- **Google Advanced Data Analytics Certificate** — *Google · July 2023*
- **ArcGIS Pro Associate 2101** — *Esri · January 2023*
- **Databases and SQL for Data Science** — *IBM · November 2023*
- **Google Project Management Certificate** — *Google · May 2023*
- **Programming with JavaScript** — *Meta · July 2023*

REFERENCES

Nicole Ueberschär	Technical Director *Esri Rwanda*	n.ueberschaer@esri.rw
Hubert Bagambiki	Chief Inspector *City of Kigali*	hbagambiki@kigalicity.gov.rw
Theogene Ntakirutimana	Watershed Protection Specialist	t.ntakirutimana@greenfund.rw